Tribhuvan University
Institute of Science and Technology
Amrit Campus

A *Lab Report* on

Design and Analysis of Algorithm (CSC-325)

Submitted to:
Department of Computer Science & Information Technology
Amrit Campus, Thamel, Kathmandu

Tribhuvan University

In partial fulfillment of the requirement for the *Bachelor's Degree in Computer Science & Information Technology*

Submitted By:

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Submitted To:

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Submission Date:

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Signature

LABWORKS

S.No.	Name of Experiments	Date of Submission	Signature	Remarks
1.	**Implementation of Iterative Algorithms** Task1: Generating Fibonacci Series. Task2: Calculating GCD using Euclidean Algorithm. Task3: Implementing Sequential Search. Task4: Implementing Bubble Sort. Task5: Implementing Selection Sort Task6: Implementing Insertion Sort.			
2.	**Implementing Divide and Conquer Algorithms** Task1: Binary Search Task2: Min-Max Finding Task3: Merge Sort Task4: Quick Sort Task5: Heap Sort			
3.	**Implementing Greedy Algorithms** Task1: Implementing Fractional Knapsack Task2: Implementing Kruskal's Algorithm to find MST Task3: Implementing Dijkstra's Algorithm Task4: Implementing Huffman Coding Algorithm			
4.	**Implementing Dynamic Programming Algorithms** Task1: Implementing Matrix Chain Multiplication to determine minimum number of scalar multiplications required. Task2: Implementing 0/1 Knapsack Problem to determine maximum profit. Task3: Implementing Floyd-Warshall Algorithm to find shortest paths between all pairs of vertices.			
5.	**Implementing Backtracking Algorithms** Task1: Implementing Subset Sum Problem. Task2: Implementing 0/1 Knapsack Problem. Task3: Implementing N-Queen Problem.			
6.	**Implementing Number Theoretic Algorithms** Task1: Implementing Chinese Remainder Theorem. Task2: Implementing Miller-Rabin Primality Test.			
7.	**Implementing Approximation Algorithms** Task1: Implementing Vertex Cover Problem. Task2: Implementing Subset Sum Problem.			